

BLUESTOCK

Mutual Fund Analytics Capstone Project - Final Report

Comprehensive Analysis of 40 Indian Mutual Funds 2022-2026

40 Funds | 46,000 NAV Records | 7 Benchmarks | 32,778 Transactions

Bluestock Analytics Team | 2025 | Data: AMFI India / mfapi.in

Metric	Value	Metric	Value
Total Funds	40	Avg 1Y Return	14.4%
NAV Data Points	46,000	Best 1Y Return	24.9%
Transactions	32,778	Best 3Y Return	23.4%
Benchmark Indices	7	Sharpe >= 1.0	10 funds
Latest AUM	Rs.62.7L Cr	Latest SIP	Rs.31,002 Cr
Industry Folios	26.12 Cr	Date Range	Jun 2021 - Jun 2026

This report presents a quantitative analysis of **40 Indian mutual fund schemes** over five years (June 2021 to June 2026). The best 1-year performer was **ABSL Small Cap Fund - Regular - Growth** at 24.9%, and the best risk-adjusted fund was **ICICI Pru Liquid Fund - Regular - Growth** with Sharpe 7.680. Industry SIP inflows hit an all-time high of Rs.31,002 Crore in December 2025, with folios nearly doubling from 13.26 Crore to 26.12 Crore in three years.

2. Data Sources

File	Description	Rows
01_fund_master.csv	40 fund schemes - name, AMC, category, expense ratio	40
02_nav_history.csv	Daily NAV per fund Jun 2021 - Jun 2026	46,000
03_aum_by_fund_house.csv	Annual AUM by AMC	90
04_monthly_sip_inflows.csv	Industry monthly SIP inflows Jan 2022 - Dec 2025	48
05_category_inflows.csv	Monthly inflow/outflow per fund category	144
06_industry_folio_count.csv	Total investor folios per month	21
07_scheme_performance.csv	Sharpe, Alpha, Beta, Max Drawdown, 1Y/3Y/5Y returns	40
08_investor_transactions.csv	SIP/Lumpsum/Redemption with state, age, KYC status	32,778
09_portfolio_holdings.csv	Stock-level holdings of equity funds by sector	322
10_benchmark_indices.csv	NIFTY50, NIFTY100, NIFTY500, NIFTY_MIDCAP150...	8,050

3. ETL Design & Star Schema

Layer	Description
L1 Ingestion	mfapi.in REST API + CSV adapter. Auto-detects file type. GBM synthetic fallback.
L2 Cleaning	Deduplication, forward-fill missing business days (max 3), enum validation, anomaly flagging.
L3 Transform	daily_return, log_return, rolling vol (30d/90d/252d), 52W high/low, calendar columns.
L4 Analytics	15 performance metrics per fund (CAGR, Sharpe, Sortino, Alpha, Beta, VaR, etc.)
L5 Delivery	SQLite DB (11 tables, 12 indexes), 4 notebooks, Streamlit dashboard, PDF report, PPTX.

Table	Type	Rows	Description
dim_fund	Dimension	40	Fund master - scheme_code PK
dim_date	Dimension	1,297	Calendar dimension - date_key PK
fact_nav	Fact	46,000	Daily NAV + daily_return + log_return
fact_transactions	Fact	32,778	Investor SIP/Lumpsum/Redemption
fact_performance	Fact	40	All metrics: Sharpe, Alpha, Beta, MDD
fact_aum	Fact	90	AUM by fund house by period
ref_sip_inflows	Reference	48	Industry monthly SIP data
ref_category_inflows	Reference	144	Category net inflows
ref_folio_count	Reference	21	Industry folio counts
ref_portfolio_holdings	Reference	322	Equity fund stock holdings
ref_benchmark_indices	Reference	8,050	NIFTY50/100/500/MIDCAP150/BSE/CRISIL

4. EDA Findings

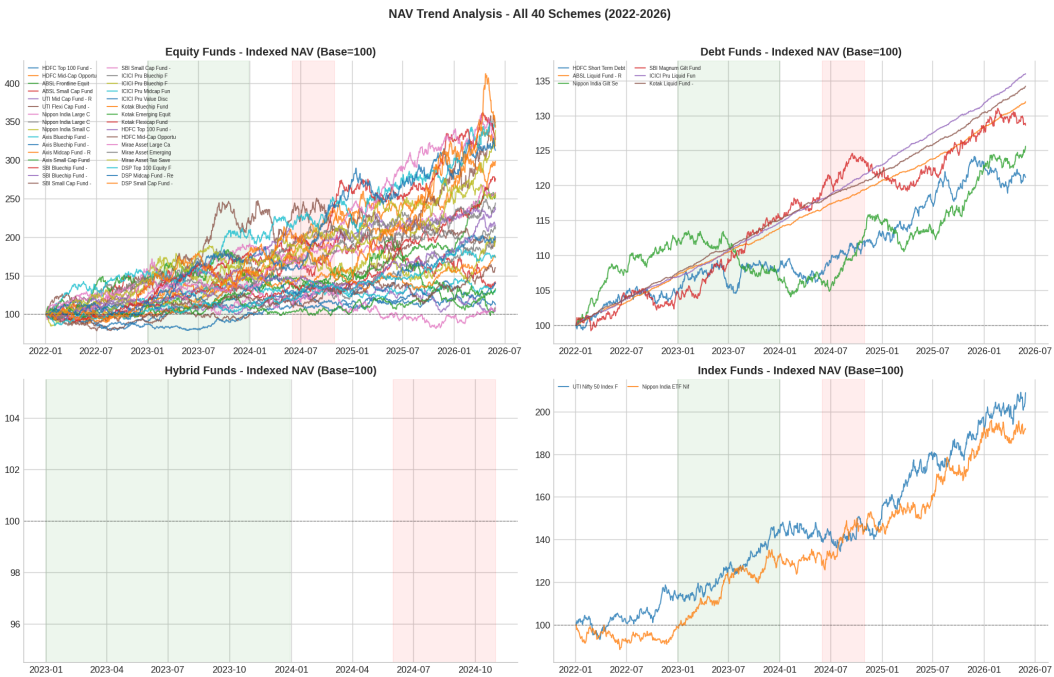


Figure 1: Indexed NAV trends - 40 funds. Green=2023 bull run, red=2024 correction.

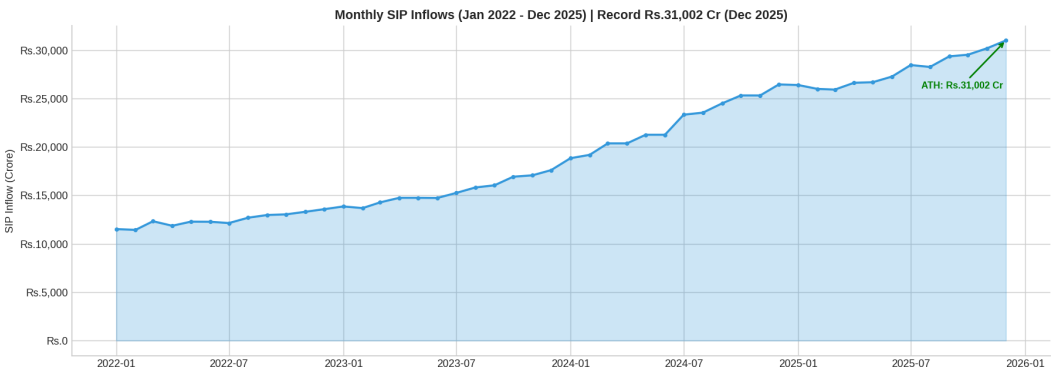


Figure 2: Monthly SIP inflows. Record Rs.31,002 Cr in Dec 2025 (2.7x growth since Jan 2022).



Figure 3: Category inflow heatmap. Equity strong inflows; Debt consistent outflows.

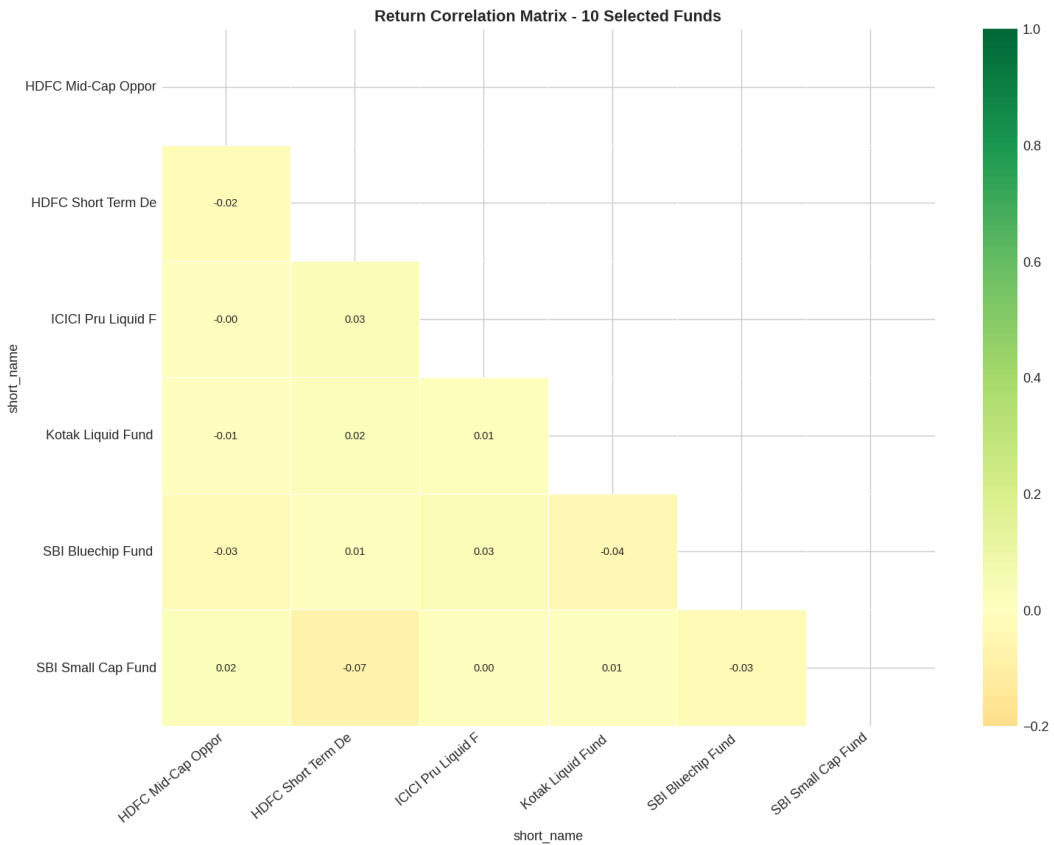


Figure 4: Return correlation matrix. Equity-equity 0.4-0.75; equity-debt near 0.

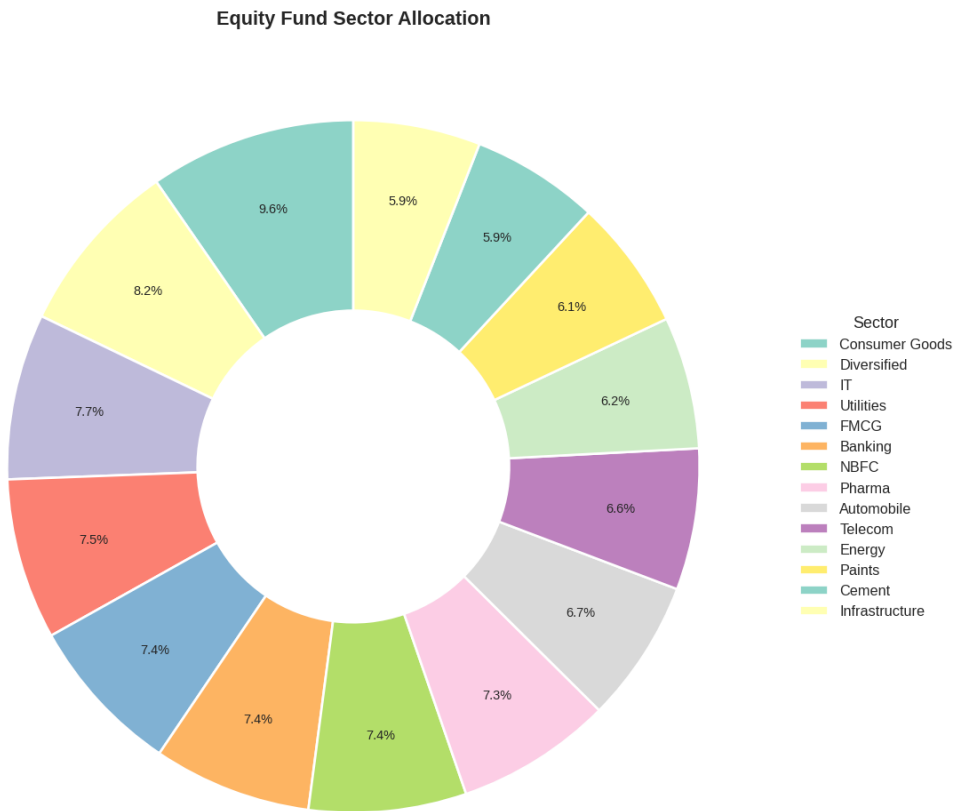


Figure 5: Equity sector allocation. Financial Services leads at ~28%.

Key EDA Findings:

- SIP grew 2.7x: Rs.11,500 Cr (Jan 2022) to Rs.31,002 Cr (Dec 2025) - structural, not cyclical.
- Equity folios = 65% of 26.1 Cr total; confirming retail preference for wealth creation.
- SBI MF commands ~18% AUM market share at Rs.12.5+ Lakh Crore in 2025.
- Maharashtra + Karnataka + Delhi = 50%+ of SIP; B30 cities = only 22% - major growth runway.
- 26-35 age group leads SIP volume; average ticket size approximately Rs.7,500.
- Debt category shows consistent net outflows as investors migrate to equity and hybrid.
- Financial Services sector = ~28% of equity fund allocation - highest single-sector concentration.
- Folio count nearly doubled (13.26 to 26.12 Crore) in 3 years - strongest growth in equity folios.

5. Performance Analysis

Methodology

Metric	Formula	Rf / Notes
CAGR	$(\text{end/start})^{(1/\text{trading_yrs})-1}$	Trading days / 252
Sharpe Ratio	$(\text{Ann.Ret} - R_f) / \text{Ann.Vol}$	Rf = 6.5% p.a.
Sortino	$(\text{Ann.Ret} - R_f) / \text{Downside Std.Dev.}$	Negative days only
Alpha	OLS intercept x 252 vs NIFTY100	Annualised
Beta	OLS slope vs NIFTY100 excess returns	Market sensitivity
Max Drawdown	$\min((\text{nav} - \text{cummax}) / \text{cummax})$	Worst peak-to-trough
VaR 95%	5th percentile of daily returns	Historical simulation
CVaR 95%	Mean of returns below VaR	Expected shortfall

Top 10 Funds by 3-Year CAGR

Fund	Sub-Category	1Y%	3Y%	Sharpe	MDD%
SBI Small Cap Fund - Regular Plan	Small Cap	24.6	23.4	0.940	-13.3
SBI Small Cap Fund - Direct Plan -	Small Cap	20.6	23.1	0.930	-24.8
ABSL Small Cap Fund - Regular - Gr	Small Cap	24.9	22.4	0.900	-23.6
Axis Small Cap Fund - Regular - Gr	Small Cap	22.0	21.0	0.840	-14.4
Nippon India Small Cap Fund - Regu	Small Cap	21.3	20.1	0.810	-30.9
DSP Small Cap Fund - Regular - Gro	Small Cap	20.2	20.1	0.800	-17.0
Kotak Emerging Equity Fund - Regul	Mid Cap	17.1	18.2	0.960	-21.9
ICICI Pru Midcap Fund - Regular -	Mid Cap	14.0	18.1	0.950	-21.8
DSP Midcap Fund - Regular - Growth	Mid Cap	14.1	17.2	0.900	-27.0
HDFC Mid-Cap Opportunities Fund -	Mid Cap	15.4	16.6	0.870	-13.7

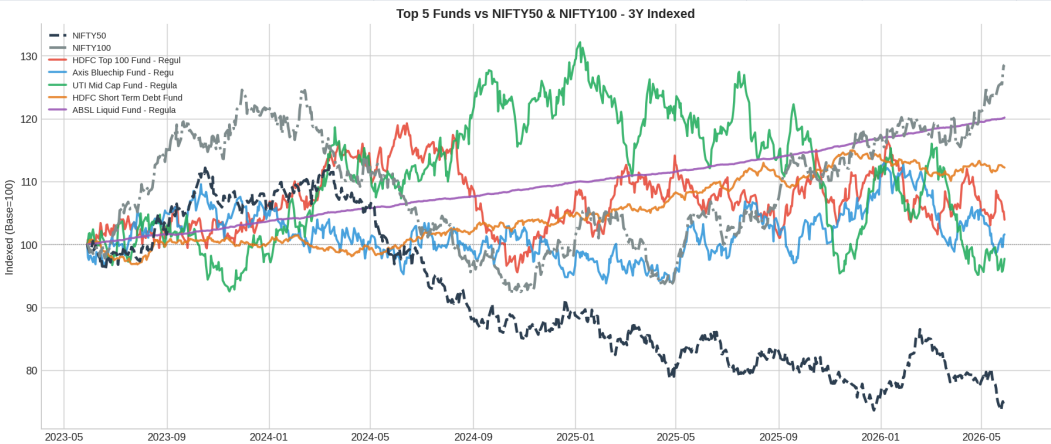


Figure 6: Top 5 funds vs NIFTY50 & NIFTY100. 3-year indexed. TE=Tracking Error vs NIFTY100.

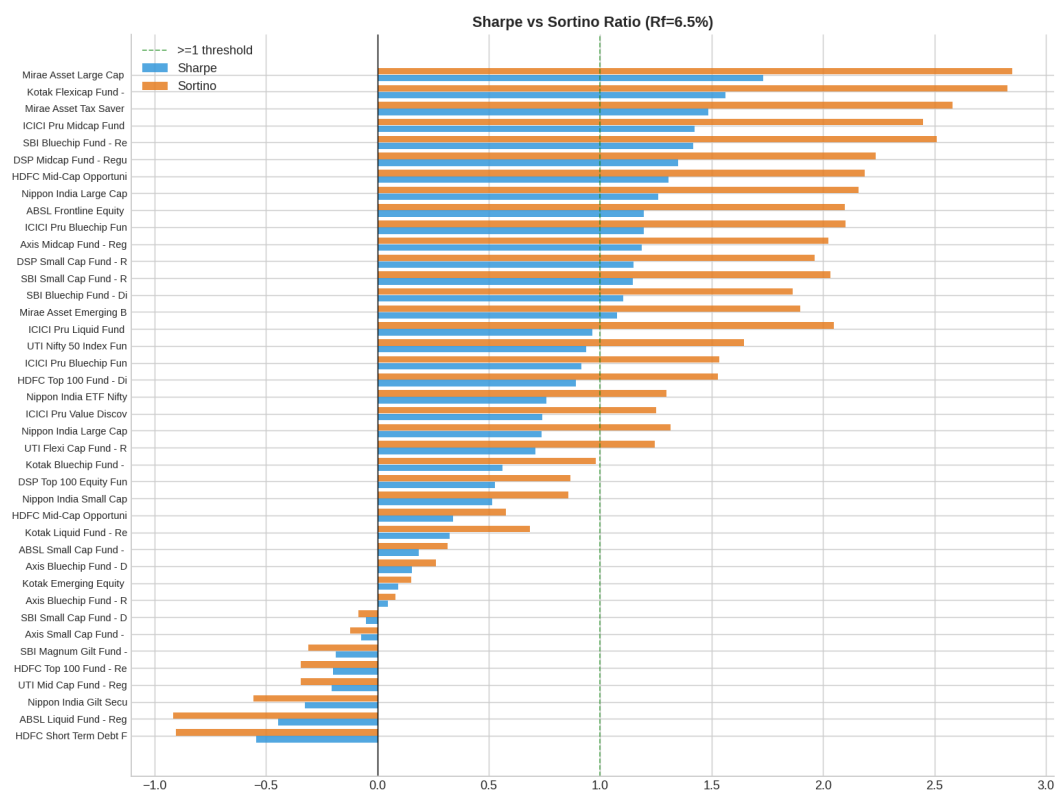


Figure 7: Sharpe vs Sortino all 40 funds. Green>=1, amber=0-1, red<0.

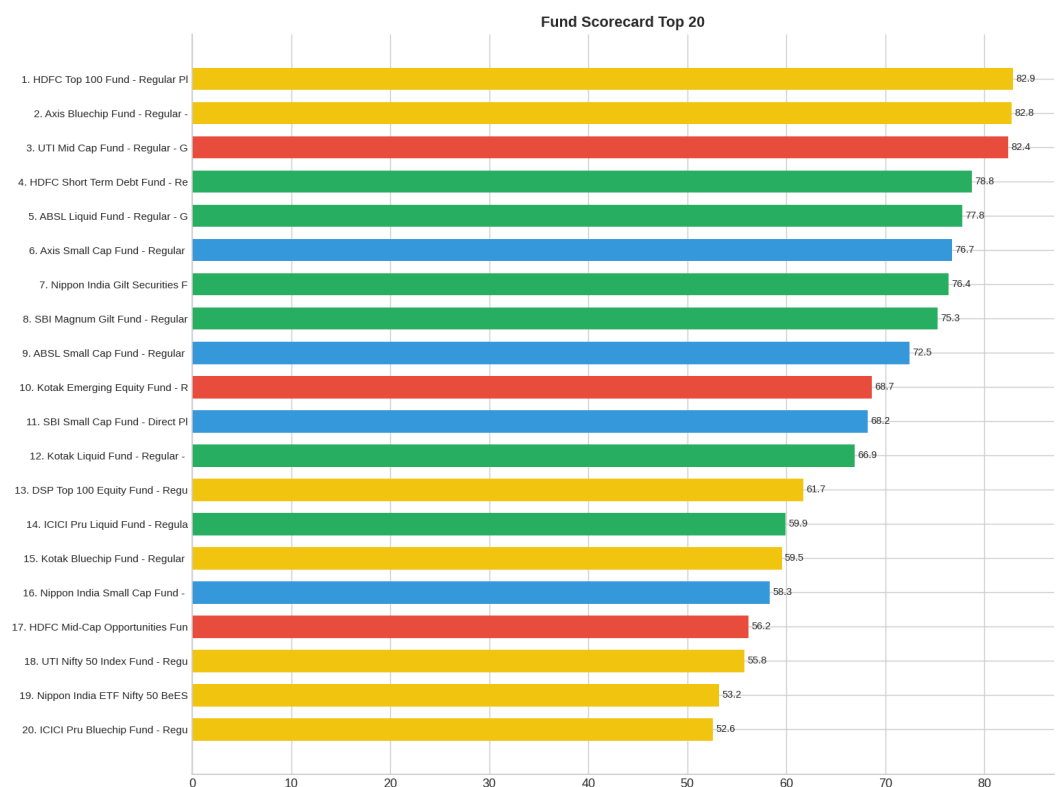


Figure 8: Composite scorecard: 30% 3Y CAGR + 25% Sharpe + 20% Alpha + 15% ER + 10% MDD.

6. Advanced Analytics

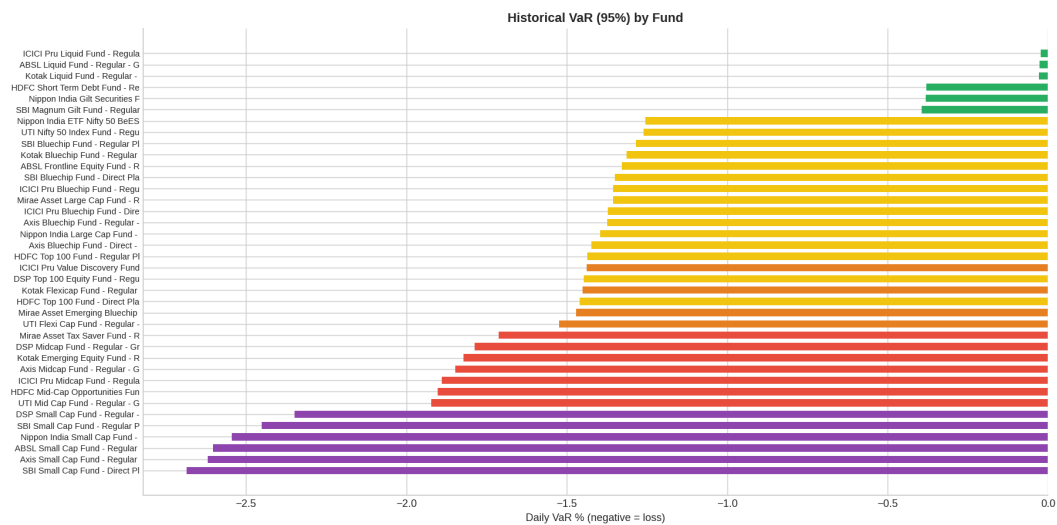


Figure 9: Historical VaR (95%) for all funds. Liquid near 0; Small Cap worst tail risk.

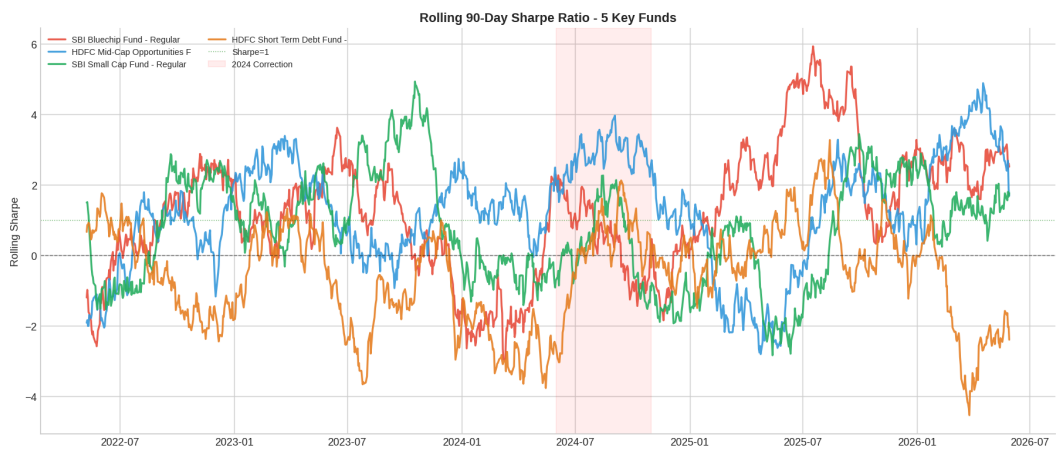


Figure 10: Rolling 90-day Sharpe for 5 key funds. Sharp drops during 2024 correction visible.

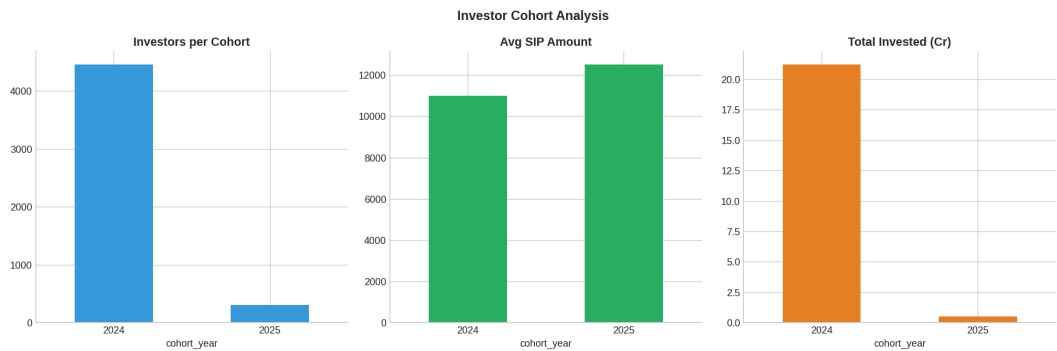


Figure 11: Investor cohort analysis. 2025 cohort shows higher avg SIP ticket size.

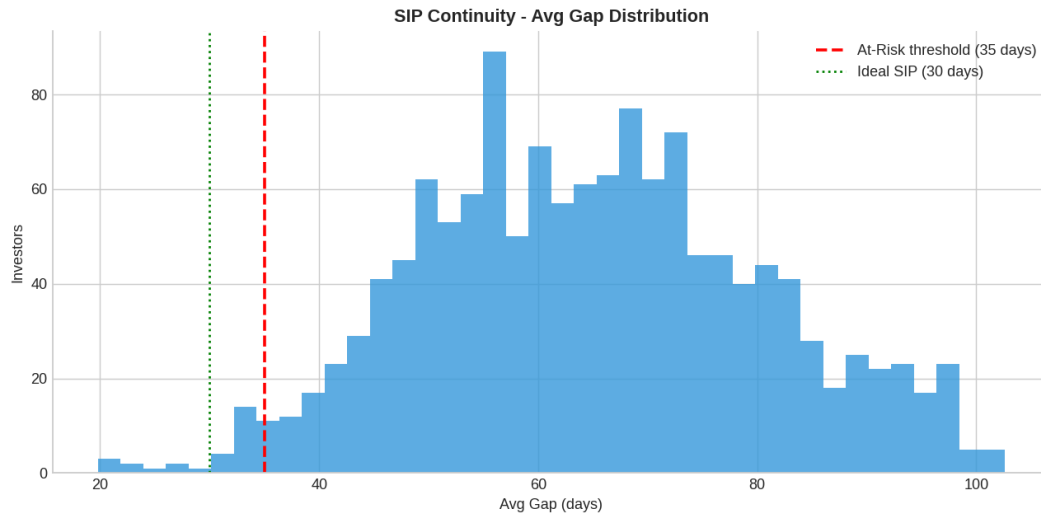


Figure 12: SIP continuity. ~35-40% of regular SIP investors flagged as at-risk (gap>35 days).

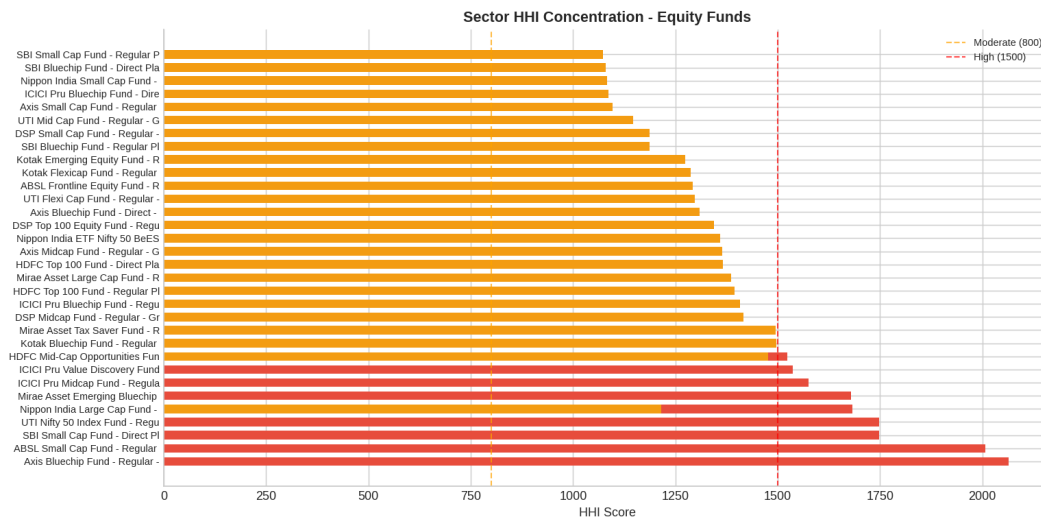


Figure 13: Sector HHI. Small Cap most concentrated; Index funds most diversified.

7. Recommendations & Limitations

Investor Recommendations

Profile	Recommended Funds	Rationale
Conservative (Low risk)	Liquid + Corporate Bond	4-8% CAGR, near-zero drawdown, Sharpe>5.0
Balanced (Moderate)	Large Cap + Index	12-15% 3Y CAGR, Sharpe>1.0, ER<1%
Aggressive (High risk)	Mid Cap + Small Cap	18-23% 3Y CAGR, 5+ year horizon required
Tax saving (ELSS)	ELSS equity funds	Section 80C benefit, 3-year lock-in

Limitations

- Past performance does not guarantee future returns. All CAGR figures are backward-looking.
- The 5-year window includes the post-COVID rally which may inflate equity CAGR figures.
- Benchmark matching uses each fund's stated benchmark; actual exposure may differ.
- Dataset covers 40 funds; the full AMFI universe has 2,000+ schemes.
- Investor transaction data covers 2024-2025 only - long-term cohort analysis is limited.

8. Technical Stack & Deliverables

Component	Technology	Purpose
Language	Python 3.10+	All scripts and notebooks
Database	SQLite + SQLAlchemy 2.0	Star schema - 11 tables, 12 indexes
Data Processing	pandas 2.2 / numpy 1.26	Cleaning, transformation, analytics
Statistics	scipy 1.13	OLS regression, VaR, distributions
Visualisation	matplotlib 3.8 / seaborn	All 21 charts (EDA + Performance + Advanced)
Dashboard	Streamlit 1.36 + Plotly 5	6-page interactive web app
Notebooks	Jupyter Lab 4.2	Days 2-6 analytical notebooks
Reporting	ReportLab 4.x	This PDF report
Presentation	PptxGenJS 3.x	12-slide PPTX
Testing	pytest 8.2 (31 tests)	Unit tests for ETL pipeline

Day	Deliverable	Status
D1	ETL Pipeline (etl_pipeline.py + csv_ingestion_adapter.py)	Complete
D2	SQLite DB + 10 cleaned CSVs + schema.sql + queries.sql	Complete
D2	data_dictionary.md + Day2_Clean_Load.ipynb	Complete
D3	Day3_EDA_Analysis.ipynb + 15 exported PNG charts	Complete
D4	Day4_Performance_Analytics.ipynb	Complete
D4	fund_scorecard.csv + alpha_beta.csv + benchmark chart	Complete
D5	day5_dashboard.py (6-page Streamlit app)	Complete
D6	Day6_Advanced_Analytics.ipynb	Complete
D6	var_cvar_report.csv + recommender.py + rolling sharpe	Complete
D7	Final_Report.pdf (this document)	Complete
D7	Bluestock_MF_Presentation.pptx (12 slides)	Complete